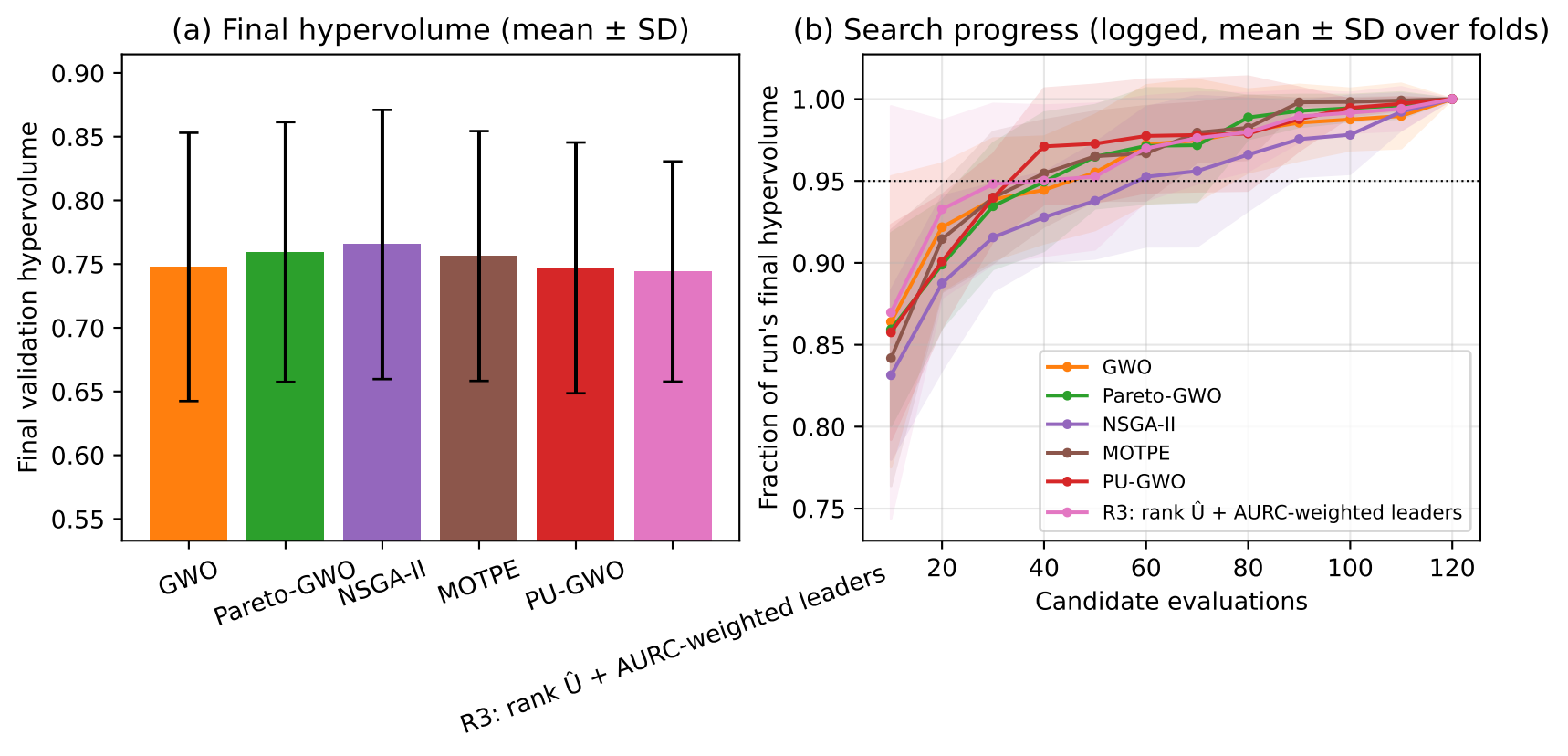
(a) Final hypervolume (mean ± SD)
Final validation hypervolume
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
GWO
Pareto-GWO
NSGA-II
MOTPE
PU-GWO
R3: rank Û + AURC-weighted leaders
(b) Search progress (logged, mean ± SD over folds)
Fraction of run's final hypervolume
1.00
0.95
0.90
0.85
0.80
0.75
20
40
60
80
100
120
Candidate evaluations
GWO
Pareto-GWO
NSGA-II
MOTPE
PU-GWO
R3: rank Û + AURC-weighted leaders